

Influence of Urban Landscape on Ants and Spiders Richness and Composition in Forests

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Abstract

As large amounts of natural environments are lost due to urbanization, the role of remnant native vegetation in the preservation of biodiversity has become even more significant. Remnant native forest patches are essential refugia for flora and fauna and are crucial for the maintenance of ecosystem processes in urbanized landscapes. We evaluated the influence of landscape structure on ants and spiders associated with Atlantic Forest remnants in urban landscapes. We sampled 14 forest areas in the Metropolitan Region of Salvador and tested the effect of the landscapes' proportion of forest cover, mean landscape isolation, and mean landscape shape complexity on the taxonomic and functional richness and the community composition of both groups. The species collected were classified into functional groups based on behavioral attributes and environmental preferences. Overall, there were strong adverse effects of forest loss, decreased connectivity, and an increase in edge effects associated with the mean shape complexity of the forest remnants. However, the spiders responded to all three landscape structure characteristics whereas the ants only responded to the landscape mean shape complexity. Our findings indicate that the maintenance of urban forest habitats is essential for the conservation of biodiversity in the Metropolitan Region of Salvador and the preservation of ecological functions performed by species within the forest areas.

Introduction

Urban environments frequently contain a mosaic of habitats with varying degrees of complexity in structure and biodiversity (Byrne 2007, Faeth *et al* 2011). A higher degree of biodiversity is found in remnants of native vegetation, e.g., forest patches and urban protected areas (Dias *et al* 2006, Pacheco & Vasconcelos 2007, Wener 2011, Nielsen *et al* 2014). These remnants provide essential refugia for native flora and fauna,

allowing for the conservation of portions of the native biodiversity and playing a vital role in maintaining rare and endangered species (Pacheco & Vasconcelos 2007, Lundholm & Richardson 2010, Wener 2011, Nielsen *et al* 2014). It is especially true for cities located in threatened ecosystems such as the Atlantic Forest, a biodiversity hotspot (Myers *et al* 2000). Therefore, understanding how biodiversity responds to changes in the landscape due to urbanization, is vital for planning biodiversity-friendly cities in such environments. In

turn, the biodiversity associated with these forest remnants performs ecological functions which are necessary for the maintenance of the forests themselves and can contribute to provisioning, supporting, regulating, and cultural ecosystem services (Ditt *et al* 2010, Scarano & Ceotto 2015, Livesley *et al* 2016, Enedino *et al* 2018). Furthermore, biological communities are more than a collection of organisms; two communities with similar metrics of taxonomic diversity (i.e., species richness, abundance, and diversity indices) however, with different functional profiles, can respond differently to deleterious environmental changes in terms of biodiversity loss and disruption of ecosystem processes (Walker 1992, Melliger *et al* 2018). In this context, the study of functional diversity can bring further insight into the consequences of urbanization on ecosystems, reaching beyond the traditional approach based on taxonomic diversity (Laureto *et al* 2015, Silva *et al* 2015).

Thus, to assess the effects of forest loss, it is vital to study organisms that have essential ecosystem functions, such as the megadiverse arthropods with crucial ecosystem functions, that can indicate drastic biodiversity losses as well as anticipate impacts to the ecosystem processes (Walker 1992, Gerlach *et al* 2013). Many studies have successfully used ants and spiders as indicators of biodiversity, with extensive literature detailing their life history, basic biology, interspecific interactions, and habitat preferences (Gerlach *et al* 2013). These studies documented the significant role of ants and spiders as ecosystem engineers and in shaping the environment by controlling arthropod populations. Previously, both groups have demonstrated sensitivity to minor disturbances such as habitat disturbances in urban settings (Alarukka *et al* 2002, Uno *et al* 2010), despite the lack of studies at the landscape level evaluating the effect of landscape structure on species richness and taxonomic and functional community composition. Ants and spiders may differ in their responses to urban landscape structure due to their biological distinctions in dispersion capacity, foraging strategies, and relationship with microclimate (Fahrig 2003, 2013, Tews *et al* 2004, Lach *et al* 2010, Foelix 2011, Melliger *et al* 2018).

For instance, there are three significant aspects of landscape structure to be considered when evaluating the role of forest remnants in the maintenance of biodiversity in urban landscapes. In general, the traditional urbanization process tends to diminish the amount of non-constructed areas, including forest remnants, reducing landscape connectivity, and increasing the shape complexity of vegetation remnants (Turner & Gardner 2015, Liu *et al* 2016). These landscape changes are mostly deleterious to the bulk of the biodiversity associated with forest remnants and are major drivers to species loss due to human activity (Fahrig 2003, 2013, Stein *et al* 2014, Turner & Gardner 2015, García-Martínez *et al* 2017). In this context, the suppression of green areas in

urban landscapes may lead to taxonomic and functional diversity loss and, consequently, jeopardize ecosystem functions and services (Schutz & Schulze 2015, García-Martínez *et al* 2017). There are mixed results in the literature and the response to landscape change is context-dependent, varying with the biological group studied and its relationship with the landscape and the methods used in the studies. This scenario makes it difficult to apply this knowledge to urban landscape management (Fahrig 2017). Despite this apparent complexity, the explanations for the varied patterns observed can be narrowed down to a handful of causal mechanisms.

On the one hand, more urbanized areas tend to have low or no resources and prohibitive conditions for the survival and reproduction of the organisms significantly associated with the forest environment. In cases such as these, the urban forest patches can practically function as habitat islands within an inhospitable urban matrix for many species, conforming to the landscapes' binary approach (Fahrig 2003, 2013). Furthermore, the Atlantic Forest urban remnants are significantly more heterogeneous than the anthropogenic environments, which associates the increase of forest cover to the rise in landscape heterogeneity (Fahrig *et al* 2011, Stein *et al* 2014, Perović *et al* 2015, Stein & Kreft 2015). Landscapes with more significant heterogeneity can maintain greater species richness due to the landscape complementation effect (Dunning *et al* 1992). The landscape complementation effect can contribute to the species richness in a landscape, since landscapes composed of multiple environments can maintain species associated with each environment and species that depend upon multiple environments (Fahrig *et al* 2011, Stein *et al* 2014).

Additionally, the amount of contact surface between forests and their surrounding environments can have distinct effects on community structures, depending on the types of associations between the species that make up communities with the forest and the surrounding environments (Fahrig 2017, Fahrig *et al* 2019). The amount of contact surface will benefit the species that can take advantage of the resources and conditions available in both the forest and surrounding environments. In this scenario, forests and surrounding environments have a complementary role in maintaining these species, rather than representing an inhospitable matrix (Fahrig 2017). On the other hand, for the species that are exclusively dependent on the resources and conditions present in the forest remnants and cannot benefit from the surrounding environments, the contact surface can have deleterious edge effects (Fahrig 2003; Ewers & Didham 2007, Lopes *et al* 2009, Uno *et al* 2010, Driscoll *et al* 2013). In this scenario, there will be a negative relationship between species richness and the amount of contact surface between the forest and the surrounding environments. In the same context, depending upon species' dispersion capacities, the distance at which the inhospitable matrix isolates forest patches

can act as a significant barrier for the exchange of organisms between patches. This process generates a myriad of deleterious effects of fragmentation, such as reducing the effective area available, disrupting metapopulation dynamics and gene flow, all of which can contribute to local extinctions and species loss (Fahrig 2003). Ultimately, the balance between the positive and negative effects of the interface between the forest and the surrounding environments may act as environmental filters, changing the community composition.

Therefore, this study aimed to evaluate the effect of urban landscape structure on ant and spider communities. More specifically, we evaluated the hypothesis that (I) the landscape proportion of Atlantic Forest cover has a positive effect on the taxonomic and functional richness of spiders and ants; (II) the mean isolation of forest patches in the landscapes has a negative impact on the taxonomic and functional richness of spiders and ants; (III) the mean shape complexity of forest patches in the landscapes has a negative effect on taxonomic and functional richness due to deleterious edge effects on the biodiversity associated with the forest remnants; and (IV) the landscape structure has an effect on the ant and spider community compositions.

Materials and Methods

Study area

The Metropolitan Region of Salvador, located in the state of Bahia–Brazil, contains 13 municipalities spread over a territory of 4337.72 km². It is the 7th largest metropolitan region of Brazil, with 3,929,209 inhabitants. Despite the size of the resident population and the level of urbanization, the Metropolitan Region of Salvador harbors many Atlantic Forest remnant patches, with the prevalence of Dense Ombrophilous Forest (SOS Mata Atlântica 2016) which are considered to be high-priority areas for the conservation of invertebrates (MMA 2002).

Sampling design

Fourteen sampling units were selected within forest patches, distributed across five municipalities (Supplementary Information Table S1 and Fig 1) in the Metropolitan Region of Salvador, all composed of Dense Ombrophilous Forest vegetation. All sampled forest patches were in reasonable shape, composed of arboreal vegetation with canopy cover, and were within the urban matrix. From July to October 2012, 10 sample points were distributed 30 m apart within each forest patch, totaling 140 sample points for the whole study. Three sampling methods were used to assess ants and spiders from both the soil and the vegetation. In the soil, leaf

litter animals were sampled with Winkler traps in 50 × 50 cm units and were kept there for 24 h, followed by animal extraction. The vegetation-dwelling fauna was sampled with an entomological umbrella (1 m²) at each sample point from three bushes (from 1 to 3 m high) shaken for 10 s. Manual diurnal capture from the soil and vegetation was also conducted for 15 min at each sample point.

Samples of ants were studied and deposited for identification at the Laboratório de Mirmecologia (CPDC, reference number #5703), Itabuna, BA (curator: J. Delabie), *Comissão Executiva do Plano da Lavoura Cacaueira* (CEPLAC). Spider specimens were sent to the Arachnological Collection of the Instituto Butantan (IBSP, curator: A.D. Brescovit), São Paulo, for identification and were deposited under number IBSP #164953. Nomenclature for ant species followed Bolton (2019), and the nomenclature for spider species followed the World Spider Catalog (2020). Species were later grouped into functional groups. For ants, features of trophic position, nest and foraging location, type of foraging, worker recruitment mode, colony size and workers, behavior during interspecific interactions, the agility of workers, and morphology were used to classify species into functional groups (nomenclature follows Silvestre *et al* 2003 and Brandão *et al* 2012). For spiders, the time of capture, foraging site, foraging mode, web use, and type of web were used to characterize species into functional groups (nomenclature follows Dias *et al* 2010). The conditions and resources used by both taxa (spiders and ants) were the criteria used for their classification into functional groups (Supplementary Information Table S2 and Supplementary Information Table S3).

Quantifying the landscape structure

Landscape composition and configuration metrics can characterize the landscape structure defined in the FRAGSTATS software (McGarigal 1995, Turner & Gardner 2015). The landscape composition metrics are not spatially explicit and, basically, express how much of each cover class there is in the landscape. In this study, the proportion of Atlantic Forest cover was adopted as representative of both the amount of habitat and compositional landscape heterogeneity of the landscapes. The latter can favor both the species that depend exclusively upon the forest for surviving and the species that can use complementary resources from the forest and the surrounding environments (Fahrig *et al* 2011, Stein & Kreft 2015). Contrary to the landscape composition, the configurational metrics are focused on the quantitative description of the spatial arrangement of land cover types. To quantify the landscape configuration, two metrics were used. The landscape mean Euclidean nearest neighbor distance of all forest patches was adopted as representative of the overall isolation of the forest patches within the

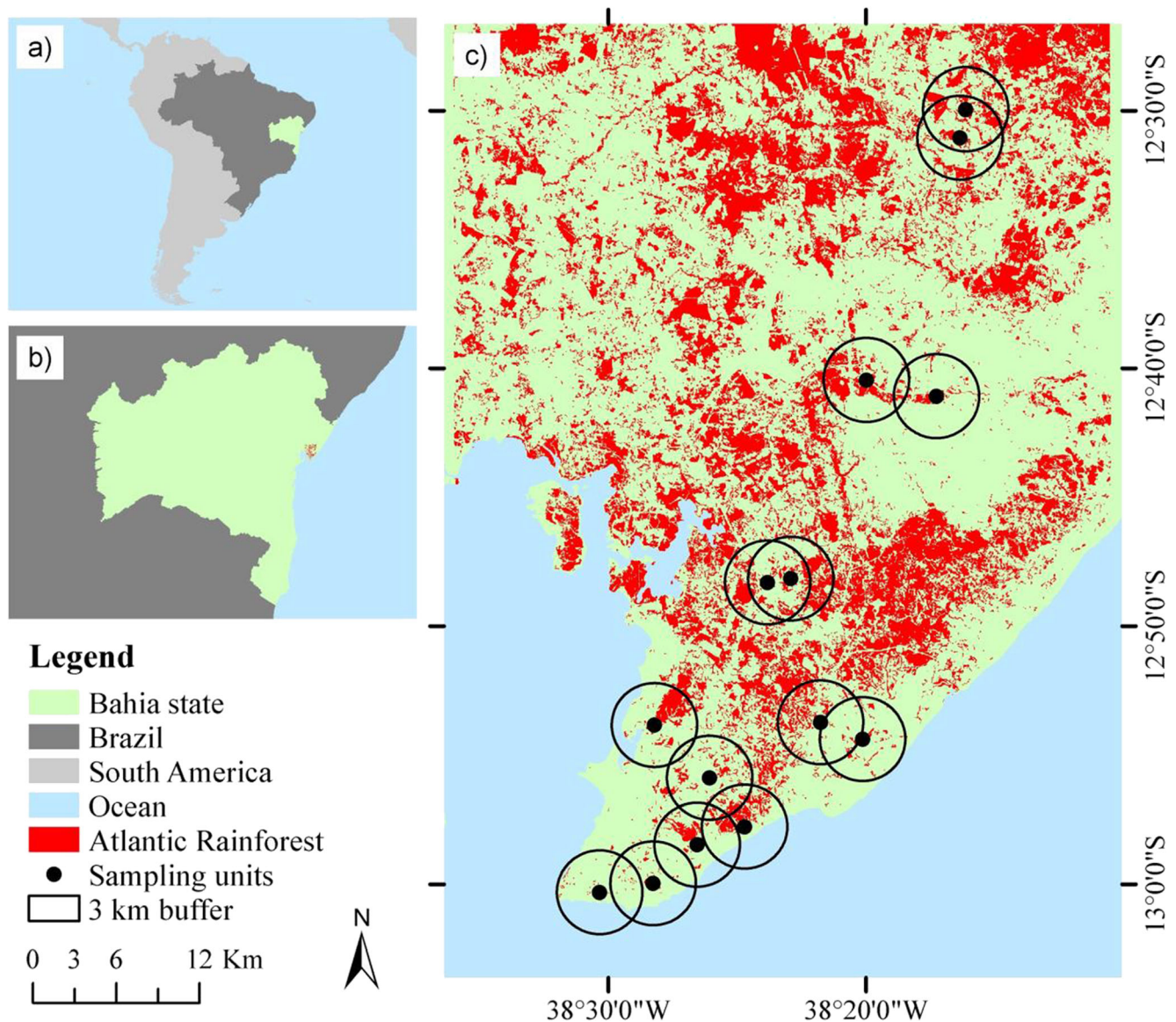


Fig 1 Distribution of all 14 forest fragments sampled within five municipalities from the Metropolitan Region of Salvador (MRS).

landscape. Furthermore, the mean fractal dimension of the shapes of all forest patches within the landscape was selected as representative of the contact surface between the land cover types. The fractal dimension is equal to two times the logarithm of patch perimeter (m), corrected for raster bias, divided by the logarithm of patch area (m^2). The fractal dimension value ranges between one and two, where values close to one mean simpler shapes (round or squares) whereas values close to two mean highly complex shapes and less compact patches, which can increase the contact surface between the forest patches and the surrounding environments. This may increase processes that depend upon the exchange of organisms, matter, and energy between these two environments. These metrics were selected because they are good descriptors of landscape patterns and because

they were not correlated with each other (Fahrig 2003, 2013, Turner & Gardner 2015). All landscape metrics were computed for each sampling unit at the landscape level using circular buffers with radii varying from 250 to 3000 m, with increments of 250 m in the R software packages *raster* version 3.3-7 and *landscapemetrics* version 1.4.4 (Hesselbarth *et al* 2019).

To measure the landscape structure, a forest cover classification of Sentinel-2 multispectral satellite images was acquired on 08/09/2016, with 10 m (Drusch *et al* 2012). Random forest supervised classification was performed based on training ground through data. After the initial classification, patches with areas lower than $4500 m^2$ were eliminated to reduce the noise created by urban afforestation. These processes achieved an Atlantic Rain Forest cover

classification with 97.8% accuracy (1% commission and 1.2 omission errors), evaluated with a testing set of complementary ground truth data that was not used in the algorithm training procedure. Manual corrections were performed to reverse forest cover changes that occurred between the sampling of ant and spider communities and the referred satellite image acquisition date. These procedures were performed using the R software packages *caret* version 6.0-86, *raster* version 3.3-7, and QGIS version 3.14.1.

Statistical analysis

To evaluate the sampling effort, a Jackknife 1 species richness estimator was used for each sampling site (Magurran 2004). It is relatively common in tropical regions that the cumulative curves of hyperdiverse groups such as ants and spider do not approximate an estimator curve nor do they stabilize, mostly due to the number of rare species in tropical environments and the variety of habitats they inhabit (Toti *et al* 2000, Willott 2001, Longino *et al* 2002). The data was randomized 999 times to remove the ordered sequence effect from samples in the Jackknife 1 procedure. These analyses were performed with the *Vegan* package for the software R-3.3.0 (Oksanen *et al* 2016).

Because of the colonial behavior prevalent between ant species (Longino 2000), to build the ant community matrix, the incidence of the samples within each sampling unit was used instead of the total number of individuals collected whereas the spider community matrix was composed of the total abundance in each sampling unit (forest patch). The functional community structure was represented by the functional groups' presence/absence matrix in each sampling unit for both ants and spiders. Both taxonomic and functional richness was calculated for ants and spiders. To analyze the variation in ant and spider community compositions, the dimensionality of the community matrices was reduced to a pair of synthetic axes using a nonmetric multidimensional scaling (NMDS) ordination technique with the Bray-Curtis index as a measure of dissimilarity. The analysis was performed separately for assemblages of ants and spiders, using the respective community matrices described above, where species abundance was normalized by the subtraction of the mean value and dividing by the standard deviation, which avoids problems with scale differences between the variations of different species. The final NMDSs can be considered great representations of both ant and spider communities, with the stress of 0.17 and 0.16, respectively (Supplementary Information Fig S1). The NMDSs were performed using the R software version 3.6.1, package *Vegan*, version 2.5-6 (Oksanen *et al* 2016).

To fine-tune the evaluation of the relationship of ant and spider communities (species richness, functional richness, and composition) with the surrounding landscape structure

(Atlantic Forest proportion, mean Euclidean nearest neighbor, and mean fractal dimension), the landscape metrics described above were computed using buffers with varying radii and the best buffer radius for each combination between a response variable and a highest R^2 criterion selected landscape metric. The buffer radii corresponding to the highest R^2 value for each analysis were then selected and used to formally test the relationship of the variables through simple linear regressions. The data for both the buffer size selection procedure and the linear regression tests were normalized through the subtraction of the mean value and dividing by the standard deviation. Normalizing the data preserved the original variation while adjusting it to comparable intervals and standardizing the slope coefficients (β). More details of the buffer size selection can be found in the support information (Supplementary Information Fig S2, S3, S4, S5, and S6; and Table S4). All calculations were performed in R version 3.6.1. We assumed the level of significance for the linear regression to be $p = 0.0167$ (0.05/3) since we did not perform alpha or p value corrections on the analyses.

To facilitate the interpretation of the results associated with the synthetic community composition variables generated via the NMDS, auxiliary exploratory analyses were performed. These analyses included different data visualization techniques, such as NMDS biplots and stacked area charts with cumulative values of the relative proportions of subfamily and functional group species richness along with the landscape structure, the analysis of the relationships between the NMDS axes with species and functional richness, and interpretation of the weighted average of species scores in the respective NMDSs. These auxiliary exploratory analyses were performed only where a significant relationship between the NMDS axes and the landscape structure was detected in the main hypothesis tests described above. All exploratory analyses were performed in R version 3.6.1 with the Package *ggplot2* version 3.3.0.9.

Results

Species richness, functional groups, and species estimators

A total of 174 species of ants were collected, distributed across eight subfamilies (Supplementary Information Table S2). Myrmicinae was the richest subfamily (98 species; 55.6% of the total), followed by Ponerinae (25; 14.2%) and Formicinae (21; 11.9%). Species were categorized into 18 functional groups, of which the most highly represented were the generalized Myrmicinae (32 species) and dominant arboreal ants associated with carbohydrate-rich resources (28). The observed richness in the forests varied from 34 to 65 species, and the estimated richness ranged from 50.2 to 96.5 (Supplementary Information Table S1). Regarding spiders,

155 species were collected, belonging to 30 different families (Supplementary Information Table S3). The richest family was Theridiidae (31 species; 20% of the total), followed by Salticidae (29; 18.7%) and Araneidae (27; 17.4%). Species were categorized into 12 functional groups, the most represented of which were orb-weavers (45 species) and diurnal space web-weavers (44). The observed richness varied from 23 to 37 species, and the estimated richness ranged from 35.6 to 61.3 (Supplementary Information Table S1).

Influence of the urban landscape on communities' composition, and taxonomic and functional richness

Ants and spiders responded differently to the landscape structures (Table 1). In general, the main predictor for ant community structure was the landscape mean fractal dimension; furthermore, there was no statistically significant response to other landscape structure features. The ant function richness was the only response variable from the ant community that was not significantly explained by any landscape characteristic.

Ant species richness was strongly negatively related to the landscape mean fractal dimension ($\beta = -0.77$; Fig 2a). Regarding ant communities' composition, variations represented by the synthetic variables NMDS1 and NMDS2 were also both strongly negatively related to the landscape mean fractal dimension (respectively $\beta = -0.88$, $\beta = -0.7$; Fig 2b and c; Supplementary Information Fig S7 a). There was no clear or apparent association between the ant NMDS1 and NMDS2, with no changes in the proportion of ant subfamilies nor functional groups (respectively Fig 3a and b; Supplementary Information Fig S7 b and c). However, the ant community composition variation, represented in the synthetic variable NMDS1, is associated with the species loss in the landscape mean fractal dimension (Fig 2), since NMDS1 is significantly related to ant species richness, to which at least 61% of NMDS1 variation can be attributed ($\beta = 0.78$, $F_{1,12} = 18.96$, $R^2 = 0.61$, $p < 0.001$). Moreover, most species with absolute weighted average species NMDS1 scores above 0.6 were positive (21 out of 28), denoting a high positive association with NMDS1 and, therefore, a negative association with the landscape mean fractal dimension (Supplementary Information Table S5). By the same criteria, the remaining seven species, i.e., *Camponotus melanoticus* Emery 1894, *Neivamyrmex gibbatus* Borgmeier 1953, *Odontomachus bauri* Emery 1892, *Odontomachus brunneus* Patton 1894, *Pheidole gr. diligens* sp.17, *Pheidole gr. fallax* sp.9, and *Paratrachymyrmex* sp.1, were negatively associated with NMDS1, and therefore positively affected by the landscape mean fractal dimension. Regarding the ant NMDS2, there was no significant taxonomic or functional association, which may indicate that NMDS2 may be associated with composition variation at the species level along the gradient

of landscape mean fractal dimension. There were at least ten species with scores close to the extremes of NMDS2 which indicates a high association with said synthetic variable, i.e., *Acanthognathus rudis* Brown & Kempf 1969, *Atta cephalotes* Linnaeus 1758, *Azteca ovaticeps* Forel 1904, *Cephalotes pusillus* (Klug 1824), *Cerapachys splendens* Borgmeier 1957, *Leptogenys bohlsi* Emery 1896, *Pheidole gr. diligens* sp.8, *Pheidole gr. diligens* sp.2, *Pheidole gr. diligens* sp.29, and *Pseudomyrmex* group *pallidus* sp.4 (Supplementary Information Table S5). All of these species were positively associated with NMDS2 and, therefore, negatively affected by the landscape mean fractal dimension.

Considering the spider communities, each response variable was related to different landscape characteristics. Only the community composition variation represented by the synthetic variable NMDS2 was not significantly related to any of the landscape metrics. The spider species richness was strongly positively related to landscape forest cover proportion ($\beta = 0.8$; Fig 4a) whereas it was negatively related to the landscape mean Euclidean nearest neighbor distance ($\beta = -0.7$; Fig 4b). Spider functional richness was negatively related to landscape mean Euclidean nearest neighbor distance ($\beta = -0.67$; Fig 4c). Regarding spider community compositions, only NMDS1 was significantly related to the landscape mean fractal dimension ($\beta = 0.72$; Fig 4d; Supplementary Information Fig S7 d). There is no clear or apparent association between the spider NMDS1 and NMDS2 with changes in the proportion of spider subfamilies nor functional groups (respectively Fig 3c and d; Supplementary Information Fig S7 e and f). Furthermore, the spider NMDS1 does not show any association with either species or functional richness. However, there were at least 22 spider species with extreme NMDS1 scores, indicating their strong association with this synthetic variable (Supplementary Information Table S5). Of those species, nine were positively associated with NMDS1 and, therefore, benefited from the landscape mean fractal dimension, whereas the remaining thirteen were negatively associated with NMDS1 and were, therefore, negatively affected by the landscape mean fractal dimension. These results imply that the variation in spider community compositions is reflecting the turnover of species along the landscape mean fractal dimension gradient.

Discussion

In general, our results show that the structure of urban landscapes has a significant influence in species richness, functional richness, and community composition. The forest loss, the amount of edge, and isolation of the forest patches had an overall negative effect on the communities studied, corroborating the theoretical expectations from a forest fragmentation framework (Fahrig 2003, 2013). Accordingly, these results suggest that for the ant and spider

Table 1 Results from the linear regressions between communities and landscape structure for ants and spiders.

Group	Variables	Slope (β)	Std. error	<i>t</i> value	R^2	<i>F</i>	<i>p</i>
Ants	Species richness						
	Prop 750 m	− 0.24	0.28	− 0.85	0.06	0.72	0.4121
	ENND 750 m	0.49	0.25	1.95	0.24	3.81	0.0748
	FD 1000 m	− 0.77	0.18	− 4.22	0.60	17.82	0.0012
	Functional richness						
	Prop 250 m	− 0.32	0.27	− 1.16	0.09	1.34	0.2701
	ENND 2250 m	− 0.31	0.27	− 1.14	0.10	1.29	0.2778
	FD 2500 m	− 0.46	0.26	− 1.80	0.21	3.24	0.0972
	Community composition NMDS1						
	Prop 2250 m	− 0.27	0.28	− 0.97	0.07	0.93	0.3537
	ENND 2500 m	0.30	0.28	1.09	0.09	1.18	0.2993
	FD 1000 m	− 0.88	0.14	− 6.41	0.77	41.10	< 0.0001
	Community composition NMDS2						
	Prop 3000 m	0.32	0.27	1.16	0.10	1.35	0.2687
	ENND 2500 m	− 0.36	0.27	− 1.32	0.13	1.75	0.2110
	FD 500 m	− 0.70	0.21	− 3.37	0.49	11.37	0.0056
Spiders	Species richness						
	Prop 2250 m	0.84	0.16	5.26	0.70	27.65	0.0002
	ENND 2250 m	− 0.70	0.21	− 3.43	0.50	11.78	0.0050
	FD 250 m	0.56	0.24	2.32	0.31	5.36	0.0391
	Functional richness						
	Prop 3000 m	0.50	0.25	2.02	0.25	4.07	0.0665
	ENND 2000 m	− 0.67	0.22	− 3.09	0.44	9.52	0.0095
	FD 1750 m	0.55	0.24	2.31	0.31	5.31	0.0399
	Community composition NMDS1						
	Prop 2000 m	0.16	0.28	0.57	0.03	0.33	0.5778
	ENND 500 m	− 0.36	0.27	− 1.33	0.13	1.77	0.2085
	FD 2500 m	0.72	0.20	3.64	0.52	13.22	0.0034
	Community composition NMDS2						
	Prop 3000 m	0.59	0.23	2.56	0.35	6.55	0.0250
	ENND 3000 m	− 0.62	0.23	− 2.71	0.38	7.33	0.0191
	FD 1750 m	0.63	0.22	2.79	0.39	7.80	0.0162

R^2 , coefficient of determination; *F*, fixation indices; *p*, *p* value; *Prop*, proportion; *ENND*, Euclidean nearest neighbor distance; *FD*, mean fractal dimension

communities, the urban forest remnants function as habitat patches or fragments immersed in an inhospitable matrix of urbanized environments. However, ants and spiders differed in their responses to landscape structure. Ant communities only were significantly affected by the edge effects, whereas spiders also responded to the amount of forest and the mean patch isolation in the landscapes. Previous studies show that ants and spiders are affected by the degree of urbanization and the structural characteristics of forest patches, responding negatively to more open areas, such as gaps in forest canopy cover, and anthropized forest fragments (Melliger *et al* 2018). Below we will explore in detail our results and offer some plausible explanations to the patterns observed and differences in the responses between ant and spider communities. Additionally, we will discuss the

possible implications of our results for the conservation of biodiversity in urban environments.

Ant responses

The variations in the ant communities in our study were related only to the shape of the forest areas in the landscape. In this study, the fractal dimension negatively influenced the species richness and community composition. This overall negative effect may be due to the increase in deleterious edge effects on the organisms associated with Atlantic Forest remnants, due to the increase in the amount of border between the forest patches and the urban matrix. These findings contradict previous studies where the shape of urban forest patches did not have significant effects on ant

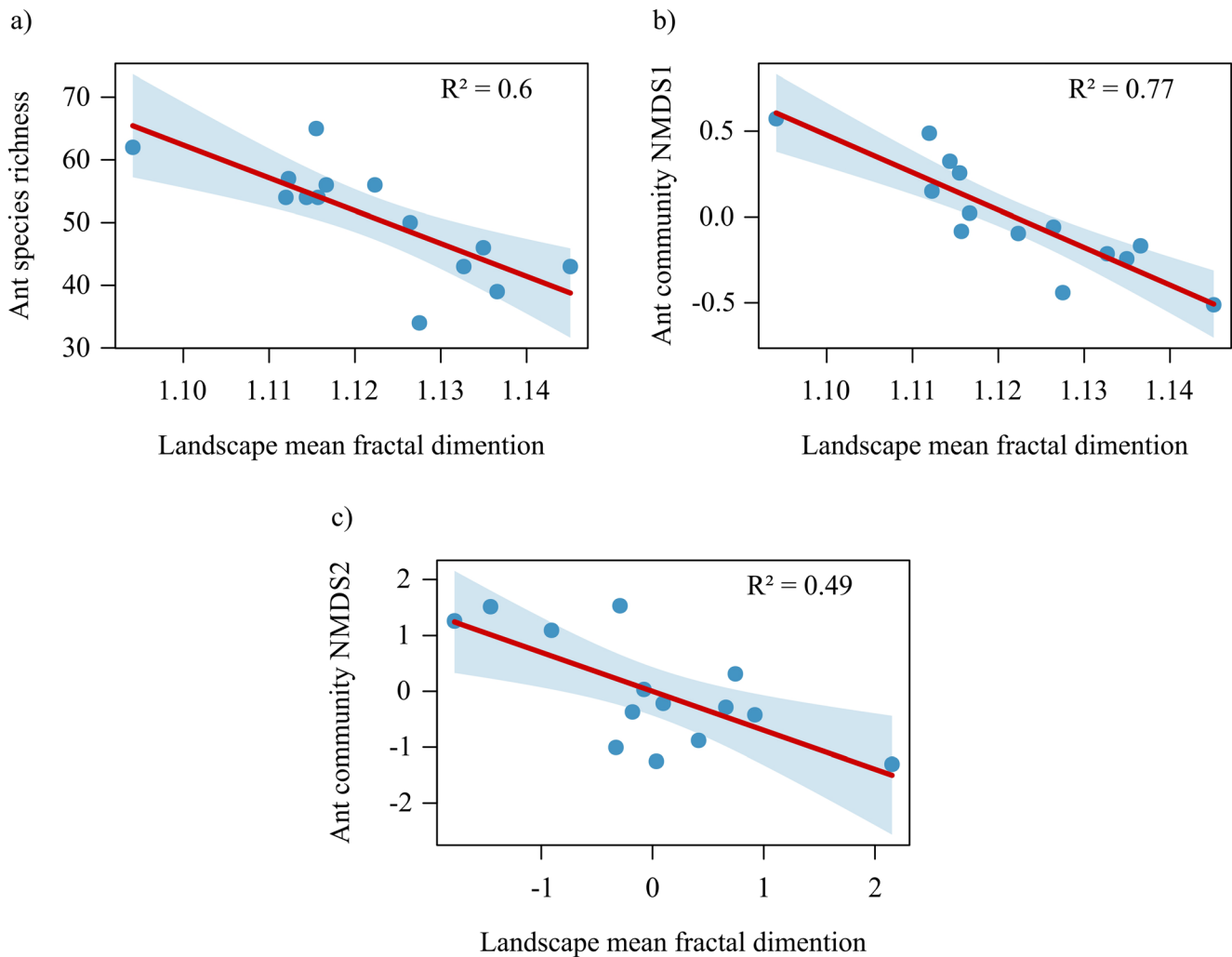


Fig 2 Linear regression results for the relationship between the ants' communities' composition, species richness, and functional groups and the landscape structure from the Metropolitan Region of Salvador; Blue points and bands represent the sampling units and 95% confidence interval respectively; red line represents the predicted values by the linear models; NMDS1 and NMDS2 refers to the nonmetric multidimensional scaling two axes.

species richness (Sobrinho & Schoereder 2007, Clarke *et al* 2008). Complex habitat fragment shapes are associated with an increase in surface contact between forest and non-forest environments and may favor groups that benefit from such transition zones to the detriment of those that do not (Sobrinho & Schoereder 2007, Ivanov & Keiper 2010, Perovic *et al* 2015, Melliger *et al* 2018).

At the edge of the forest patches, the predominant vegetation was composed of a few species of dominant pioneer heliophile plants (trees and herbs), with the absence of species from late succession stages, which contrasts with the hyperdiverse floristic composition characteristic of preserved late succession Atlantic Forest which is still present in core areas of some urban forest patches (Lopes *et al* 2009). As a result, the forest edges have more sun incidence, higher temperature, lower humidity, and less complementarity in trophic and nesting resources, compared to the interior

forest, which can have profound implications on ant community compositions (Lopes *et al* 2009, Uno *et al* 2010). For instance, the forest patch edges can favor a greater proportion of ants with aggressive and generalist behaviors (Sobrinho & Schoereder 2007, Ivanov & Keiper 2010, Melliger *et al* 2018), such as *N. gibbatus*, *O. bauri*, *O. brunneus*, *Pheidole gr. diligens* sp.17, and *Pheidole gr. fallax* sp.9, which were positively associated with the forest shape in our study. These ants can be strong resource competitors in edge environmental conditions, increasing the competitive pressure on other ant species that are more strongly associated with the forest interior, ultimately causing their competitive exclusion and diminishing species richness. Moreover, arboreal and/or cryptic behavior may help to explain the negative association of many species such as *A. rudis*, *A. cephalotes*, *Azteca ovaticiceps* Forel 1904, *C. pusillus*, *C. splendens*, *L. bohlsi*, *Pheidole gr. diligens* sp.8,

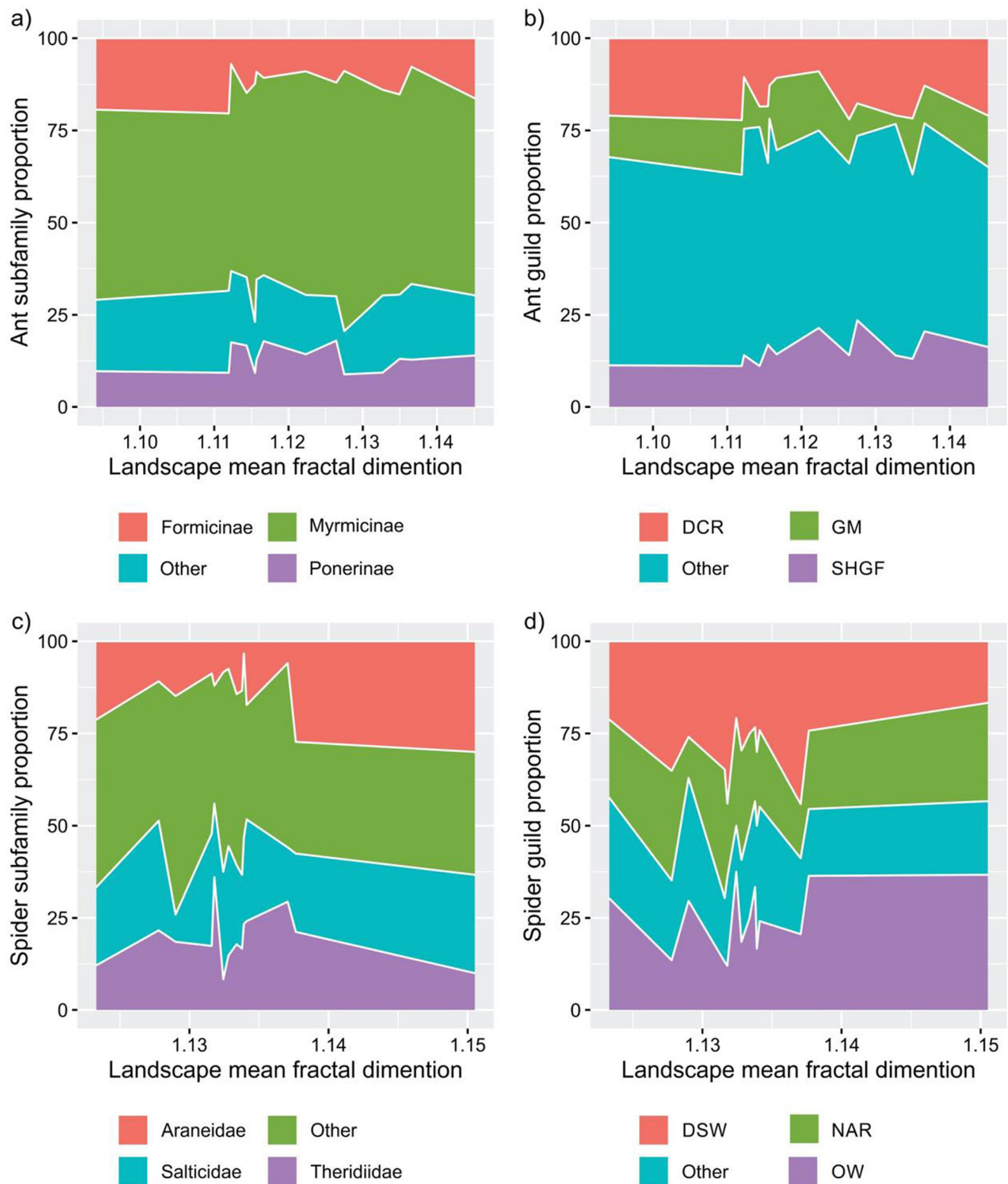


Fig 3 Community composition variation along landscape configuration gradient; relative proportion of ants' species richness of subfamilies (a) and functional groups (b) in relation to the landscape mean fractal dimension; DCR, dominant arboreal ants associated with carbohydrate-rich resources; GM, generalized myrmicines; SHGF, small-sized hypogaecic generalist foragers; for the Other groups, see information in Supplementary Information Table S2; relative proportion of spiders' species richness of subfamilies (c) and functional groups (d) in relation to the landscape mean fractal dimension; DSW, diurnal space web-weavers; NAR, nocturnal aerial runners; OW, orb weavers; for the Other groups, see information in Supplementary Information Table S3.

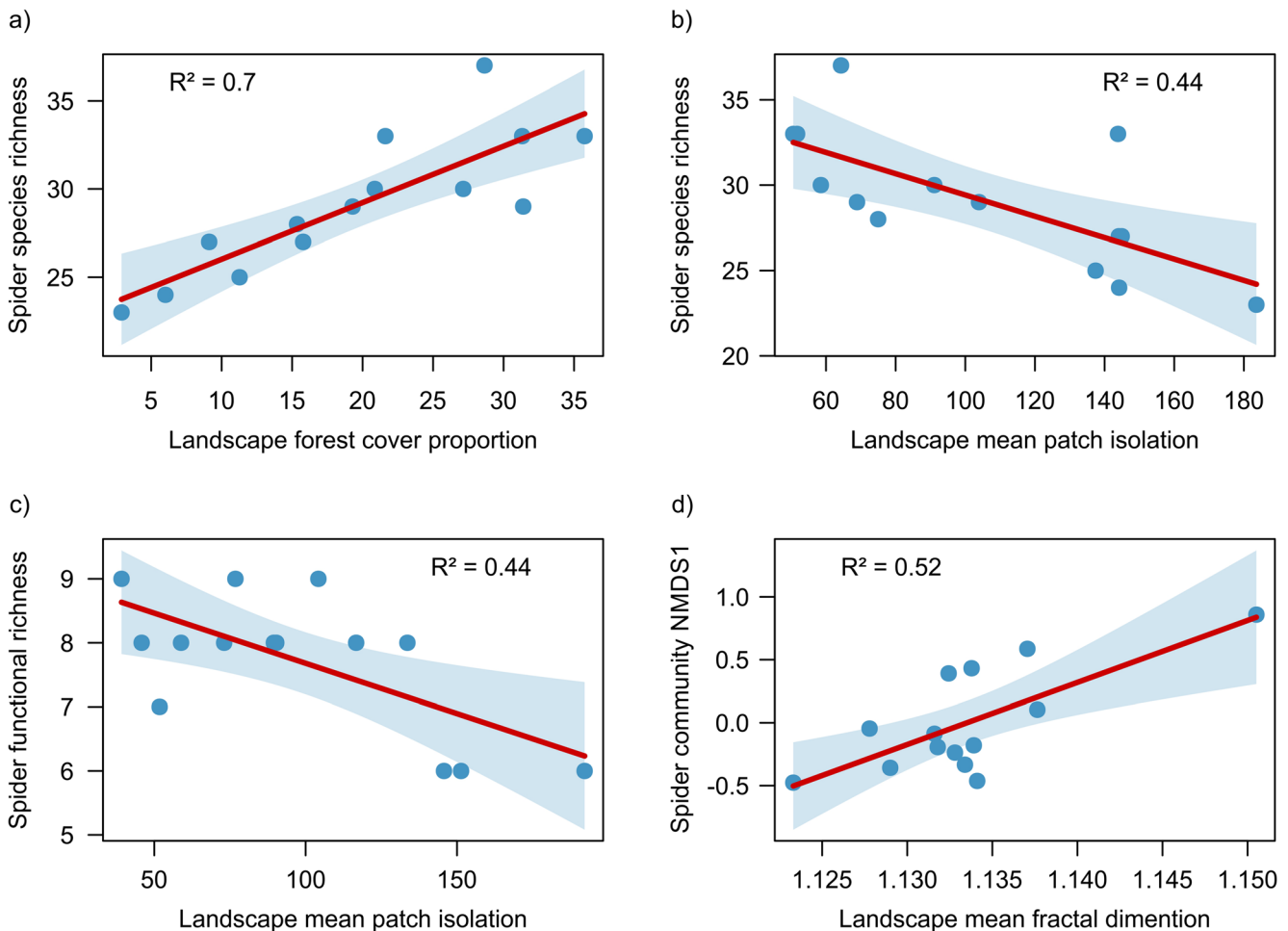


Fig. 4 Linear regression results for the relationship between the spiders' communities' composition, species richness, and functional groups and the landscape structure from the Metropolitan Region of Salvador; Blue points and bands represent the sampling units and 95% confidence interval respectively; red line represents the predicted values by the linear models; NMDS1 refers to the nonmetric multidimensional scaling first axis.

Pheidole gr. diligens sp.2, *Pheidole gr. diligens* sp.29, and *Pseudomyrmex group pallidus* sp.4 with the forest edge (Ivanov & Keiper 2010, Menke *et al* 2011, Rocha-Ortega & Castaño-Meneses 2015). These species found in the interior of the forest are mainly related to closed canopy environments (Melliger *et al* 2018), which maintains greater microclimate stability, as well as late succession vegetation and greater environmental structural complexity. Thus, the shape of the urban forest fragments can influence the ant community through changes in environmental conditions favoring generalist species that increase the competitive pressure on forest specialists that are negatively affected by edge effects, changing community compositions and reducing species richness.

Spider responses

The response of the study spider communities was very congruent with what is expected within habitat loss and

fragmentation frameworks (Fahrig 2003, 2013). In contrast to the results for ant community, spider species richness decreased with the increase of Atlantic Forest loss and isolation of forest patches. Similar results have been reported for other groups of arthropods in tropical forests (Larsen *et al* 2008, Filgueiras *et al* 2011, Storck-Tonon & Peres 2017, Salomão *et al* 2018). Previous studies show that the richness of spider species is negatively affected by urbanization in general, mainly by groups of web weavers, as they are more sensitive to changes in habitat structure (vegetation) and higher temperatures which occur in the urban matrix, when compared to soil species (Argañaraz *et al* 2017, Lövei *et al* 2018). Our study shows that this is true at the landscape level as well.

Forest loss has multiple implications for the communities associated with it, such as a diminution of carrying capacity, which reduces the resident populations, thereby diminishing its viability and increasing the probability of local extinctions (Fahrig 2003, 2013). Another implication is the reduction of

environmental heterogeneity within the forest patches. The Atlantic Forest is known for its high tree diversity, whereas in large continuous primary forest patches, each tree may be tens of kilometers away from its closest intraspecific neighbor, let alone the other vegetational strata (Lopes et al 2009). This vegetational diversity generates a huge variety of resources and conditions and, consequently, allows very complex niche partitioning and the coexistence of a very diverse fauna associated with these environments (Tews et al 2004, Stein et al 2014, Stein & Kreft 2015).

The detrimental effects of the isolation between forest patches within landscapes may also be associated with multiple processes. Firstly, travel between forest patches becomes less likely with the increase of distance and with the degree matrix impermeability (Ewers & Didham 2007, Driscoll et al 2013). For instance, most spiders (e.g., Araneidae and Theridiidae) perform ballooning, a dispersal method directly dependent on the wind that allows the animals to move for many kilometers across the landscape (Foelix 2011). Even though ballooning spiders have a greater dispersion capacity between forest patches, in urban matrices the level of verticalization may play a role in the permeability of the matrix, since tall buildings can act as fiscal obstacles to these spiders by blocking or altering the course of the wind (Argañaraz & Gleiser 2017, Lövei et al 2018). Which brings us to the second factor, even if the organisms can cross the distances between forest patches, the matrix can also increase the mortality rate in the landscape, due to physiological stress induced by poor environmental conditions, lack of resources, and exposure to predators, negatively impacting the populations (Ewers & Didham 2007, Driscoll et al 2013). This can be especially dramatic for organisms with erratic movements and with little control of their traveling course or with low distance viewing relative to the distance between forest patches, all of which may apply to spiders in general. Lastly, these obstacles in the exchange of organisms between forest patches may reduce the effective area available and also disrupt possible metapopulation dynamics, ultimately reducing the viability of these populations in the landscape (Ewers & Didham 2007, Driscoll et al 2013). All of these effects may help to explain the negative effect of the level of isolation in landscapes on the taxonomic and functional richness in our results.

Similar to the ants, the spider community compositions were also affected by the irregular shapes of the forest patches. Once more, the altered environmental conditions in the forest edges play an important role, favoring generalist species which are tolerant to the forest/matrix interface and impairing species dependent on or species more associated with the forest interior conditions (Ewers & Didham 2007, Lopes et al 2009, Uno et al 2010, Driscoll et al 2013). This is no different for spiders; the increase in complexity creates more transition borders between forests and the city, where

generalist spiders that are adapted to open environments can move between the urban matrix and the remnants of the Atlantic Forest (Magura et al 2010). In our study, some species of wandering spiders were indeed positively associated with the forest patch edge complexity, while the ambush and weaver spiders, deemed as more sensitive to matrix conditions, were negatively associated with the edge complexity of the forest patches in the landscapes (Lopes et al 2009, Uno et al 2010, Argañaraz & Gleiser 2017, Lövei et al 2018). Variations in response to fragmentation, such as these, are widely reported in the literature and may help to explain the edge effect on the spider communities detected here (Ewers and Didham 2007, Driscoll et al 2013, Fahrig 2017, Fahrig et al 2019, Merckx et al 2019).

Considerations to the differences between ants and spiders

The differences between ants and spiders found here can be at least partially explained by the specificities of their life history. For instance, spiders are exclusively predators and their distribution can be associated with the way in which they capture prey as well as the prey available, while ants tend to be generalists and feed on a wide variety of both plant and animal organic matter and, therefore, find resources with more ease (Lach et al 2010, Foelix 2011). Furthermore, the social behavior prevalent in ant colonies allows for the gathering of resources from a relatively wide area, which may make them less sensitive to changes in forest cover compared to spiders, which are almost entirely solitary (Lach et al 2010, Foelix 2011). Previous studies show that predatory species are more sensitive to the urbanization process (Corcos et al 2019), causing spiders to be influenced by more factors related to the landscape structure. Furthermore, the differences in ant and spider responses to landscape connectivity may be due to their dispersal strategies (Ewers & Didham 2007, Driscoll et al 2013). Many species of ants reproduce through mating flights which can be actively controlled by these insects, allowing them to steer the flight more directly from one forest patch to another and to more effectively avoid obstacles such as tall buildings, while the ballooning spiders passively use the wind (Lach et al 2010, Foelix 2011). This distinction may have implications for the effectiveness of dispersal movements as well as for the level of permeability of the urban matrix.

Our results show that in general, habitat loss and fragmentation negatively affected ant and spider communities. More specifically, urban landscapes with more forest cover, patches with more regular shapes and connectivity, favored the maintenance of species specialized for forest environments and richer communities of ants and spiders. Therefore, the urban Atlantic Forest remnants are crucial to the maintenance of the ant and spider diversity, as well as the ecosystem processes and services that they provide.

However, there are some differences in the responses of ants and spiders to landscape structure, which may be related to the biology of the groups. Nevertheless, ant and spider community compositions respond similarly to the forest patch edges. At the edges, both species of ants and spiders that have a more general habit in obtaining resources and are better adapted to environments more exposed to microclimate variations, were favored. Thus, we highlight the need to consider different taxonomic groups and metrics of the community when assessing the impact of urbanization on organisms, allowing for better planning in conservation actions for urban forest fragments.

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Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

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